

RISCHAN MAFRUR



WSU Lecturer's ID: 30071294

School of Computer, Data and Mathematical Sciences

Research Interest	Data Science, Artificial Intelligence, Data Warehousing, Financial Data Engineering, Blockchain and Decentralized Finance (DeFi) Analytics
Year Join WSUI as Lecturer	January 2026
Email	R.Mafrur@westernsydney.edu.au

rischanlab.github.io linkedin.com/in/rischan github.com/rischanlab kaggle.com/rischan

PERSONAL PROFILE

Biography

Dr. Rischan Mafrur is a lecturer at Western Sydney University Indonesia (WSUI), Surabaya, where he teaches Data Science and Big Data Applications. He earned his PhD in Data Science from The University of Queensland, Australia. He is also a data professional with over 10 years of experience, specialising in Python, SQL, and data warehousing. His research and applied work sit at the intersection of financial data engineering, AI systems, and digital asset analytics, supported by active collaborations across Australia and Southeast Asia.

Before joining WSUI, he worked as a researcher at Macquarie University, Australia, collaborating with financial data specialists on Australian Securities Exchange (ASX) and digital asset data pipelines. His work included extracting raw data from ASX, processing corporate actions and pricing data, and producing end-of-day datasets for academic and research use. He also contributed to the development of a crypto data warehouse that ingests DeFi data and transforms it into analysis-ready tables for academic research.

In addition, he has delivered sessions for an advanced data analytics upskilling program hosted by UNSW Sydney (Mentem), supporting professionals from Commonwealth Bank (CBA) and the NSW Government across diverse technical and non-technical backgrounds. He also works as a data scientist on a casual project basis, undertaking multiple client-requested projects across data analysis, automation, and applied analytics. Across these roles, much of his work has been carried out remotely while living a nomadic lifestyle with his family.

EDUCATION

Doctoral

The University of Queensland, Brisbane, Australia

Doctor of Philosophy in Data Science

PhD thesis: Recommending Data Visualizations

January 2018 to November 2023

Master

Chonnam National University, Gwangju, South Korea

Master of Engineering in Internet of Things

September 2013 to August 2015

Bachelor

Sunan Kalijaga State Islamic University, Yogyakarta, Indonesia

Bachelor of Computer in Informatic Engineering

August 2009 to January 2013

ACADEMIC EXPERIENCES

Lecturer

Western Sydney University Indonesia (WSUI), Surabaya, Indonesia (On-site)

January 2026 to present (Full-time)

- COMP 2025 Introduction to Data Science
- COMP 3002 Applications of Big Data
- COMP 2019 Systems Programming 1

Instructor

UNSW (Mentem) Advanced Data Analytics Upskilling, Sydney, Australia (Remote)

January 2025 to December 2025 (Part-time)

Teaching Assistant

The University of Queensland, Brisbane, Australia (On-site)

February 2020 to July 2023 (Part-time)

- Advanced Database Systems (INFS3200)
- Database Principles (INFS7901)
- Relational Database Systems (INFS2200)

Teaching Assistant

Sunan Kalijaga State Islamic University, Yogyakarta, Indonesia (On-site)

September 2010 to April 2013 (Part-time)

- Algorithms and Data Structures (C) (2010)
- Operating Systems (Linux Bash Scripting) (2010, 2012)
- Visual Programming (Borland Delphi) (2011)
- Database Systems (MySQL and Oracle) (2011)

- Structured Programming (C, Procedural) (2011, 2012)
- Object-Oriented Programming (Java) (2012)
- Web Programming (PHP) (2012)
- Information Systems (2012/2013)
- Mobile Computing (Android) (2012/2013)

PROFESSIONAL EXPERIENCES

Financial Data Engineer and Researcher

Department of Applied Finance, Macquarie University, Sydney, Australia (On Site)
December 2024 to December 2025 (Full-time)

- End-of-Day ASX Data Service: Designed automated pipelines to clean, enrich, and merge multi-source data for end-of-day reporting and time-series forecasting
- Ethereum mempool data analysis: Analysed and reported Ethereum mempool data for blockchain transaction insights.
- Centralised exchange (CEX) crypto data analysis: Analysed and reported data from centralised exchanges to assess data quality.

Financial Data Engineer

SIRCA Ltd, Sydney, Australia (Remote)
February 2023 to December 2024 (Full-time)

- Crypto data warehouse: Built a crypto data warehouse and ingested DeFi data for research data collection.
- Working on product definition project, which is instrumental in defining the data available within the data warehouse collections hosted on both AWS and our on-premises rack servers. This website can be viewed here: cryptodatawarehouse.org.

Data Consultant

Ministry of Education and Culture (MoEC), Indonesia (Remote)
July 2021 to July 2022 (Part-time)

- belajar.id: The Ministry of Education and Culture (MoEC) Republic Indonesia launched a national account program in 2019, in which the ministry provided a national email account for 27 million students as well as 3.4 million teachers. As a data consultant, I have to achieve the expected output: report of the findings, include (but not limited to) the research methodology, data exploration and observations, and recommendations on how to increase the activation and adoption rate

Founder, SME Analytics Solutions

Visua.id (Acquired), Jakarta, Indonesia
2020 to 2021

- Business intelligence delivery: Built plug-and-play dashboards across Shopee, Tokopedia, and Facebook Ads.
- Template design: Enabled scalable self-service reporting through repeatable frameworks.

Backend and Database Engineer

Kartoza Pty Ltd, Western Cape, South Africa (Remote)

September 2015 to February 2017 (Full-time)

- **Projecta:** Open-source visual changelogs for software releases and software project management.
- **AgriTechnovation:** Farm mapping and management platform for agricultural land in South Africa.
- **SANSA Catalogue:** Online Earth observation catalogue for the South African National Space Agency.

PROGRAMMING / TECH SKILLS

Programming Languages: Python (expert in Pandas), SQL, R

Databases: PostgreSQL, MySQL, Oracle

Operating Systems: Unix/Linux, macOS, Windows

Cloud: AWS, GCP

Tools: Jupyter Notebook, Docker, Git

Blockchain: Ethereum, Solana

Visualisation: Matplotlib, Seaborn, Plotly

ML Tools: scikit-learn, SciPy

RECENT PUBLICATIONS

- **Rischan Mafrur**, “Tokenize everything, but can you sell it? RWA liquidity challenges and the road ahead”, CoRR (preprint), abs/2508.11651, 2025.
- **Rischan Mafrur** and Khadijah Hasim, “From Gold Dinar to Digital Ledger: Is Islam Ready for Cryptocurrency?”, SSRN (preprint), 2025.
- **Rischan Mafrur**, “AI-Based Crypto Tokens: The Illusion of Decentralized AI?”, *IET Blockchain*, vol. 5, no. 1, Art. e70015, 2025.
- **Rischan Mafrur**, “Blockchain Data Analytics: Review and Challenges”, CoRR (preprint), abs/2503.09165, 2025.
- **Rischan Mafrur**, Mohamed A. Sharaf, and Guido Zuccon, “VizPut: Insight-Aware Imputation of Incomplete Data for Visualization Recommendation”, CoRR (preprint), abs/2311.07926, 2023.
- **Rischan Mafrur**, “Recommending Data Visualizations: Tackling Diversification and Data Quality Challenges”, PhD Thesis, School of Electrical Engineering and Computer Science, The University of Queensland, 2023.
- **Rischan Mafrur**, Mohamed A. Sharaf and Guido Zuccon, “Efficient Diversification for Recommending Aggregate Data Visualizations”, IEEE Access, 2023.
- **Rischan Mafrur**, Mohamed A. Sharaf and Guido Zuccon, “Quality Matters: Understanding the Impact of Incomplete Data on Visualization Recommendation”, *DEXA 2020*, Bratislava, Slovakia, 2020.

- **Rischan Mafrur**, Mohamed A. Sharaf and Hina A. Khan, “DiVE: Diversifying View Recommendation for Visual Data Exploration”, *CIKM 2018*, Torino, Italy, 2018.
- **Rischan Mafrur**, I Gde Dharma Nugraha and Deokjai Choi, “Modeling and discovering human behavior from smartphone sensing life-log data for identification purpose”, *International Journal of Human-centric Computing and Information Sciences*, 2015.

OTHER ENGAGEMENTS

Book Author

- **24 Purnama di Negeri Ginseng** (Author)
Motivational book for Indonesians who want to pursue education overseas, especially in South Korea. The book is available in Gramedia and online marketplaces such as Shopee and Tokopedia. Link: <https://books.google.co.id/books?id=nsj-DwAAQBAJ>

As Speaker

- Panel Speaker, IABC Networking Iftar: “**Business Unusual: AI, Climate and the New Landscape**”, organised by the Indonesia Australia Business Council (IABC) with Investment NSW, KINETIK, and Sweef Capital, 24 February 2026, The Summit Club, World Trade Centre 1, Jakarta, Indonesia.
- Speaker, Focus Group Discussion on “**Non-Communicable Disease and Machine Learning**”, conducted by the Statistical Modeling in Life Science Research Group, Faculty of Science and Technology, Airlangga University, 5 February 2026, The Southern Hotel, Surabaya, Indonesia.
- Speaker, International Public Lecture: “**Using AI to Enhance Research Productivity and Quality**”, Magister Informatika, Faculty of Science and Technology, Universitas Islam Negeri Sunan Kalijaga Yogyakarta, 27 October 2025, Yogyakarta, Indonesia (hybrid).
- Speaker, Kuliah Umum: “**Challenges and Opportunities in Remote Working**”, Informatika, Fakultas Sains dan Teknologi, UIN Sunan Kalijaga Yogyakarta, 27 October 2025, Yogyakarta, Indonesia.
- Speaker, Lite Bites 53 (Niagahoster): “**Machine Learning: What It Is and Why It Matters**”, 14 July 2022 (online, Zoom).
- Speaker, Webinar by Schoters: “**Siapkan Kariermu di Bidang Metaverse, AI, dan Blockchain!**”, 13 March 2022 (online).
- Speaker, Indonesia AI Talks Eps. 5: “**AI & Blockchain for Our Future**” (“When Artificial Intelligence Meets Blockchain Technology”), Indonesia AI, 6 March 2022 (online).
- Speaker, Digital Talk: “**Potensi Pemanfaatan Blockchain untuk Platform Crowdfunding: Challenges & Possibilities**”, organised by the Digital Department Group (Indonesia Dermawan network), 29 December 2021 (online, Zoom).

- Speaker, **“Blockchain Use Cases in Telco Industry”**, Community of Practice 2021 (WarMO / Warga PMO), Project & Implementation Directorate, Tower Bersama Group, 15 December 2021 (online, Zoom).
- Speaker, Webinar Mata Garuda Kalimantan Timur: **“Dilema Cryptocurrency Setelah Fatwa MUI”**, Mata Garuda Kalimantan Timur, 4 December 2021 (online, Zoom).
- Speaker, Pembekalan & Pelepasan Wisuda, Fakultas Sains dan Teknologi, UIN Sunan Kalijaga Yogyakarta: **“Kiat Sukses Studi di Luar Negeri dan Membangun Jiwa Kreatif di Era Digital”**, 14 June 2021, Yogyakarta, Indonesia (hybrid).
- Speaker, Guest speaker (The Leaders): **“Founder of Visua”**, hosted by Mahendra Wijaya, 14 February 2021 (live streaming, YouTube).
- Speaker, Webinar / panel discussion: **“Lulus S1 Lanjut S2 atau Kerja?”**, OUAC 3.0, 5 February 2021, 1.30 PM–4 PM (online, Zoom).
- Speaker, Lite discussion / webinar by Potensia: **“Beasiswa di Korea dan Australia: Bincang Santai tentang Navigasi Dunia Pendidikan”**, 31 January 2021 (online).
- Speaker, Seminar Pendidikan dan Kebudayaan: **“Di Korea dan Australia”**, Himpunan Mahasiswa Pendidikan Agama Islam, Universitas Alma Ata, 19 November 2020 (online, Google Meet).
- Speaker, Webinar (HMTI, Politeknik Negeri Cilacap): **“Pemanfaatan AI untuk Teknologi Informasi dan Aplikasinya”**, 10 November 2020 (online).
- Speaker, Mini Conference, Indonesian Students Association of Queensland University of Technology (ISAQ): **“From Data to Insight: Automating Data Visualization through Recommendation”**, 18 October 2020 (online).
- Speaker, Webinar AI (Bisa AI): **“From Data to Insights: Automating Data Visualization Through Recommendation”**, 10 October 2020 (online).
- Speaker, The 31st International Conference on Database and Expert Systems Applications (**DEXA 2020**), 14–17 September 2020, Bratislava, Slovakia.
- Speaker, Kajian Rutin Kamis Sore (HIMMPAS UGM): **“Membentuk Karakter Muslim Sejati untuk Menyambut Bulan yang Suci”**, 16 April 2020 (online, Zoom).
- Speaker, Seminar Online: **“Rahasia Beasiswa Korea”**, Seputarkuliah.com, 24 November 2019, 19.00 WIB (online).
- Panel Speaker, Seminar Informatika: **“IT Product Development Strategies”** (Data Science session), Teknik Informatika, UIN Sunan Kalijaga Yogyakarta, 28 March 2018, Gedung Convention Hall Lt 1, UIN Sunan Kalijaga Yogyakarta, Indonesia.
- Speaker, New Super SchoTalk #1 (SBC Bogor): **“Kupas Tuntas Study & Scholarship di Universitas TOP Asian Series: Singapur, Taiwan, Korea & Japan”**, Kelas Korea session, 23 September 2016, 19.30–21.30 WIB (online via WhatsApp).
- Speaker, Seminar Beasiswa dan Motivasi: **“Berjuang Demi Negeri dengan Prestasi”**, UIN Sunan Kalijaga Yogyakarta, 14 March 2016, Teatrikal Fakultas Sains & Teknologi UIN Suka Yogyakarta, Indonesia.
- Speaker, Seminar: **“Raih Impian Kuliah di Negeri Ginseng”**, Rumah Komunitas Ilmu Berbagi Foundation, 20 March 2016, Indonesia.
- Speaker, Islamic talk session: **“Menangkan Ramadhan di Tengah Pandemi”**, 2020 (online).

- Speaker, Life Learner Me session: **“How to Write A Good Motivation Letter”**, 7 May 2020, 15.30–16.30 WIB.
- Speaker, book discussion session: **“Bedah Buku: 24 Purnama di Negeri Ginseng”**, organised by Perpustakaan and Gerakan Literasi Sekolah SMANSA Temanggung, 26 March 2018, SMA Negeri 1 Temanggung, Temanggung, Indonesia.
- Speaker, book discussion session: **“Bedah Buku: 24 Purnama di Negeri Ginseng”**, organised by Program Pendampingan Keagamaan (PPK), Faculty of Science and Technology, UIN Sunan Kalijaga Yogyakarta, 28 March 2018, Ruang Teatrikal Fakultas Sains dan Teknologi UIN Sunan Kalijaga Yogyakarta, Yogyakarta, Indonesia.
- Speaker, workshop session: **“Android Geek Workshop”**, held on 4 May 2013, 8am–3pm, Laboratorium Terpadu, UIN Sunan Kalijaga, Yogyakarta, Indonesia.
- Speaker, Angkringan IT (bincang-bincang dan diskusi ringan seputar IT): **“Strategy of Developing Mobile Applications”**, 12 April 2013, Gedung Imagine IT Education Center, Yogyakarta, Indonesia.

Selected Links and Activities

- Kaggle Competition Expert: kaggle.com/rischan (Ranked 2,921 out of 205,757 globally). Top 3% (Silver Medal) in the Jane Street Market Prediction Competition; Top 4% (Silver Medal) in the Cornell Birdcall Identification Competition.
- Dune Blockchain Analytics: dune.com/0xgembus (dashboards primarily focused on Solana analytics using SQL).
- Writing: medium.com/@diteliti (crypto research and analytics articles).
- CryptoCoinHalal: cryptocoinhalal.com (website built to improve financial literacy for Indonesians on the halal status of cryptocurrencies).
